Chapter 10: One- and Two-Sample Tests of Hypotheses (Part I)

ENGR 0021

Learning Objectives

- Understand one-sided confidence bounds
- Understand statistical hypotheses and their components
- Learn how hypothesis tests are constructed
- Understand Type I and Type II errors
- Understand the use of p-values for decision making in hypothesis testing
- Learn single-sample tests concerning a single mean

Statistical Hypotheses

A **statistical hypothesis** is an assertion or conjecture about one or more populations.

Null hypothesis:

$$H_0$$

The claim initially assumed to be true, typically representing **chance variation** (no real effect).

Alternative hypothesis:

$$H_1 \text{ or } H_a$$

The statement that contradicts the null hypothesis, representing a **real difference or effect**.

Possible conclusions:

- Reject H_0
- Fail to reject H_0

To perform a hypothesis test, H_0 must be expressed as a **population model** for the data.

Forms of Hypotheses

The null hypothesis usually has the form

$$H_0 : \theta = \theta_0$$

The alternative hypothesis can take three forms:

Upper-tailed test

$$H_1 : \theta > \theta_0$$

Lower-tailed test

$$H_1 : \theta < \theta_0$$

Two-tailed test

$$H_1 : \theta \neq \theta_0$$

Example: Mean Hypothesis

Suppose we are interested in the mean compressive strength of concrete.

$$H_0 : \mu = 2500 \text{ psi}$$

$$H_1 : \mu \neq 2500 \text{ psi}$$

We collect sample data and determine whether there is enough evidence to reject H_0.

Example: Variance Hypothesis

Suppose we are studying the variability of a filling machine.

$$H_0 : \sigma^2 = 0.02$$

$$H_1 : \sigma^2 < 0.02$$

This is a **lower-tailed test**.

Example: Proportion Hypothesis

Suppose we monitor the fraction of defective devices.

$$H_0 : p = 0.05$$

$$H_1 : p > 0.05$$

This is an **upper-tailed test**.

Example: Loaded Die?

A die is rolled 100 times. The total of the observed face values is 365 instead of the expected 350. Question: can this be explained by **chance variation** alone, or does it suggest that the die is **loaded**?

For a fair die, the population model is

$$1, 2, 3, 4, 5, 6$$

The standard deviation of one die roll is approximately

$$\sigma \approx 1.71$$

Therefore, the standard deviation of the sum of 100 rolls is approximately

$$\sigma_{\text{sum}} = \sqrt{100}(1.71) = 17.1$$

Since

$$365 - 350 = 15 < 17.1$$

the result can reasonably be explained by chance variation.

Example: ESP Experiment

Charles Tart conducted an experiment using a machine called the "Aquarius," which randomly selected one of 4 targets. Subjects attempted to guess the selected target. Fifteen subjects each made 500 guesses, for a total of

$$15 \times 500 = 7500 \text{ guesses}$$

Out of these, 2006 guesses were correct. If subjects had no ESP ability, the probability of a correct guess would be $p = \frac{1}{4}$ so the expected number of correct guesses would be

$$\frac{1}{4}(7500) = 1875$$

This gives a surplus of

$$2006 - 1875 = 131$$

correct guesses. Is this large enough to dismiss as chance?

Testing a Statistical Hypothesis

A hypothesis test uses sample data to decide whether the null hypothesis should be rejected.

A test procedure is based on:

1. **Test statistic**
   A function of the sample data.

2. **Rejection region**
   The set of values for which H_0 will be rejected.

Errors in Hypothesis Testing

Two types of errors are possible:

Type I Error
Rejecting H_0 when it is actually true.

$$P(\text{Type I Error}) = \alpha$$

α is called the **significance level**.

Type II Error
Failing to reject H_0 when it is false.

$$P(\text{Type II Error}) = \beta$$

Decision Outcomes

	H_0 true	H_0 false
Do not reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

Example: Legal Analogy

$$H_0 : \text{Defendant is innocent}$$

$$H_1 : \text{Defendant is guilty}$$

- Convicting an innocent person = Type I error
- Letting a guilty person go free = Type II error

Decision	**Reality**	
	Defendant Innocent	**Defendant Guilty**
Acquit	Correct Decision	Error (Type II)
Convict	Error (Type I)	Correct Decision

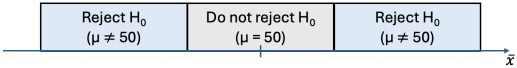
Example: Hypothesis Test for Mean

Consider the hypothesis

$$H_0 : \mu = 50$$

$$H_1 : \mu \neq 50$$

We reject the null hypothesis if the sample mean is far from 50.



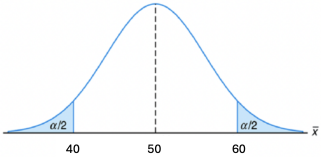
Calculating α (Type I Error)

Assume we flip a coin $n = 100$ times. Let X = number of heads.

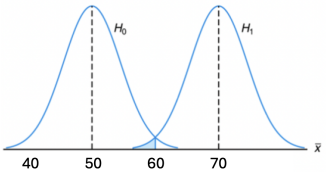
$$H_0 : p = 0.5 \quad \text{(fair coin)}$$

Suppose we reject H_0 if $X \leq 40 \quad \text{or} \quad X \geq 60$
Let the critical region be denoted by C_α.

$$\begin{aligned}
\alpha &= P(\text{reject } H_0 \mid H_0 \text{ true}) \\
&= P(X \in C_\alpha \mid p = 0.5) \\
&= P(X \leq 40 \text{ or } X \geq 60 \mid p = 0.5).
\end{aligned}$$



Calculating β (Type II Error)



Suppose the true mean is

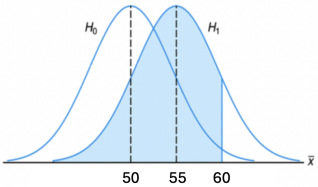
$$\mu = 70$$

Then

$$\beta = P(\text{accept } H_0 \mid \mu = 70)$$

$$\beta = P(40 < \bar{X} < 60 \mid \mu = 70)$$

Calculating β (Type II Error)



Suppose the true mean is

$$\mu = 55$$

Then

$$\beta = P(\text{accept } H_0 \mid \mu = 55)$$

$$\beta = P(67 < \bar{X} < 69 \mid \mu = 55)$$

β is now much larger.

Important Properties of Hypothesis Tests

1. Reducing α usually increases β.
2. Changing the critical value adjusts α.
3. Increasing sample size reduces both α and β.
4. β decreases as the true parameter moves further from the hypothesized value.

Power of a Test

The **power** of a test is the probability of correctly rejecting a false null hypothesis.

$$\text{Power} = 1 - \beta$$

Higher power means the test is more effective at detecting real differences.

p-value

The **p-value** is the smallest significance level at which the observed test statistic would lead to rejection of H_0.

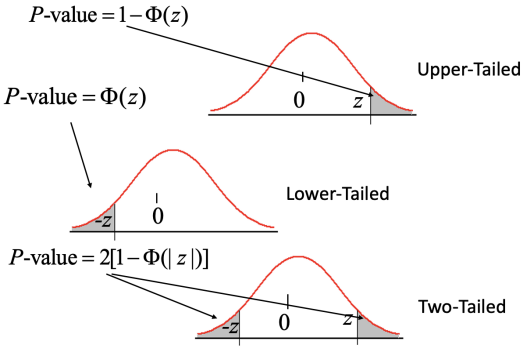
Decision rule:

- If $p < \alpha$, reject H_0
- If $p \geq \alpha$, fail to reject H_0

p-value for a z-Test

$$p = \begin{cases} 1 - \Phi(z) & \text{upper-tailed test} \\ \Phi(z) & \text{lower-tailed test} \\ 2[1 - \Phi(|z|)] & \text{two-tailed test} \end{cases}$$

A Graphical Demonstration of a P-value



Concept Question: Interpreting p-values

Other things being equal, which of the following p-values is best for the null hypothesis?

$$0.1\%, \quad 1\%, \quad 3\%, \quad 17\%, \quad 32\%$$

Repeat the question for the alternative hypothesis.

p-value Approach

Steps:

1. State H_0 and H_1
2. Choose significance level α
3. Compute the test statistic
4. Compute the p-value
5. Compare the p-value with α
6. Draw conclusions

Choosing rejection regions

Coin with probability of heads θ.

Test statistic $x =$ number of heads in 10 tosses.

$$H_0: \text{ "the coin is fair," i.e. } \theta = 0.5$$

$$H_1: \text{ "the coin is biased," i.e. } \theta \neq 0.5$$

Two strategies:

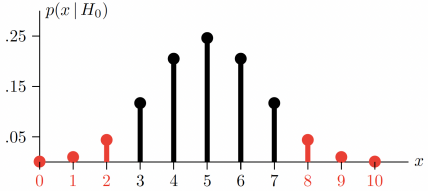
1. Choose rejection region, then compute significance level.
2. Choose significance level, then determine rejection region.

Everything is computed assuming H_0.

Concept Question

Suppose we have the coin from the previous slide.

1. The rejection region is bordered in red. What is the significance level α?



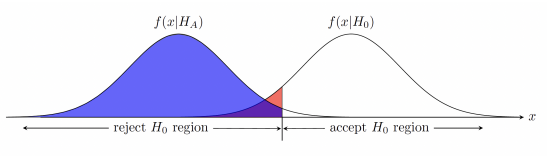
x	0	1	2	3	4	5	6	7	8	9	10
$p(x\|H_0)$	.001	.010	.044	.117	.205	.246	.205	.117	.044	.010	.001

2. Given significance level $\alpha = 0.05$, find a two-sided rejection region.

Solution

Concept question

The null and alternative pdfs are shown.



The significance level α is the area of which region?
Choices: (1) red (2) purple (3) blue (4) white (5) blue + purple
(6) red + purple (7) white + red + purple

Answer: 6. red + purple

Explanation: α is the area under the H_0 distribution that lies in the rejection region.

Example: Is the Coin Fair?

A coin is tossed 100 times. Observed: 61 heads

H_0: $p = 0.5$
H_1: $p > 0.5$

Example: Interpreting the p-value

Example: ESP Experiment

Charles Tart conducted an experiment using a machine called the "Aquarius," which randomly selected one of 4 targets. Subjects attempted to guess the selected target. Fifteen subjects each made 500 guesses, for a total of

$$15 \times 500 = 7500 \text{ guesses}$$

Out of these, 2006 guesses were correct. If subjects had no ESP ability, the probability of a correct guess would be $p = \frac{1}{4}$ so the expected number of correct guesses would be

$$\frac{1}{4}(7500) = 1875$$

This gives a surplus of

$$2006 - 1875 = 131$$

correct guesses. Is this large enough to dismiss as chance?

Interpretation

Key Takeaways: Hypothesis Testing

- **Start with a null hypothesis** (H_0): a model of chance
- **Test statistic:** summarizes how far data are from H_0
- **P-value:**

  Probability of observing data this extreme (or more) if H_0 is true

- **Small p-value** $\Rightarrow$ evidence against H_0
- **Important distinctions:**
  - Statistical significance $\neq$ practical importance
  - Statistical significance $\neq$ proof that H_1 is true
- **Always ask:**
  - Is the model (null hypothesis) reasonable?
  - Could there be bias or alternative explanations?